

Disentangling Event Importance and Link Adaptation for IoT Event Delivery: A Simulation and ESP32-S3/E220 Study

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Abstract

Periodic IoT traffic includes events whose loss matters more than routine-sample loss, but uniform redundancy ignores event value. We evaluate frozen EVENTGUARD-v1, which selects up to three proactive DATA copies using predicted event importance and first-copy ACK history. We separate three delivery objectives—CRITICAL, IMPORTANT, and overall delivery—and compare event-aware allocation with exact-budget blind controls and an importance-only ablation. The evidence comprises 8,000 host runs, 11,000 host-only sensitivity runs, and a post-hoc balanced analysis of 96 completed ESP32-S3/E220-400T22D runs from the earliest six contiguous complete seeds after a planned 160-run experiment was interrupted; all 96 passed offline audit. EVENTGUARD directed more of an identical DATA-copy budget to critical samples than blind controls, with small critical-delivery gains in some independent-copy-loss conditions and none under constructed sample-wide erasure. EVENTGUARD and IMPORTANCE-ONLY delivered the same CRITICAL events in matched host and selected hardware runs: correctly classified CRITICAL samples already receive the three-copy cap in every link state. Under software-injected RANDOM_COPY loss, EVENTGUARD showed higher IMPORTANT and overall delivery than IMPORTANCE-ONLY in selected hardware runs, while spending more copies and bytes. These six-seed device findings are descriptive, and injected loss is not measured RF packet error. The value of the evaluated link response depends on the delivery KPI, available policy action space, loss structure, and communication cost.

Keywords: constrained IoT communication; event importance; proactive redundancy; exact-budget evaluation; ESP32-S3; Sub-GHz radio.

1 Introduction

Periodic sensing is often evaluated by average packet delivery, which gives the same weight to a routine observation and a rare event that may demand intervention. In a constrained IoT link, sending additional copies of every sample consumes communication capacity; the number of copies alone says little about whether scarce transmissions reached the samples that matter. Event importance can guide this allocation. Recent ACK outcomes can also motivate adaptation to an estimated link state. These are distinct mechanisms, however, and a combined policy can obscure which one explains an observed delivery gain. We ask whether importance improves CRITICAL delivery at equal copy budgets; whether link adaptation adds delivery beyond importance alone, and for which KPI; how any gain trades against communication cost; and how programmed loss structure changes these answers.

The distinction matters especially when an importance rule already assigns the maximum allowed copies to a critical sample. A subsequent link-state transition may increase traffic for other classes while having no remaining action on the class used by the primary reliability metric. Because the frozen policy gives different link-adaptive action space to different importance classes, CRITICAL delivery alone can hide benefits and costs affecting IMPORTANT and routine traffic. Without an importance-only ablation, the resulting benefit could be attributed incorrectly to link adaptation. Without a matched budget, a combined policy’s higher delivery could instead reflect higher spending. We therefore compare EVENTGUARD with IMPORTANCE-ONLY as a non-equal-cost component ablation, and separately with event-blind allocators that receive EVENTGUARD’s **exact total DATA-copy budget** for each seed-specific trace and loss condition. Independent-copy and constructed sample-wide erasures test how the structure of loss changes the value of within-sample redundancy.

The study combines a frozen Python/C policy, an 8,000-run host matrix on evaluation seeds 31–130, 11,000 host-only sensitivity runs, and execution through a two-device ESP32-S3/E220 DATA/ACK path. The original 160-run device experiment was interrupted by an unresolved receive-path stall. We analyze the earliest six contiguous seeds with complete four-strategy/four-condition coverage: a post-hoc balanced prefix selected by execution order and completeness, not by comparative outcome. Its $n = 6$ sample size was not preregistered, so the device results support descriptive mechanism analysis rather than confirmatory inference.

This paper contributes:

1. A frozen, interpretable redundancy policy combining predicted event importance with first-copy ACK history, together with an explicit account of its action-space limits.
2. Paired deterministic evaluation and exact-budget event-blind controls that separate allocation quality from extra communication expenditure.
3. Importance-only and link-only ablations showing that importance explains frozen v1’s CRITICAL-delivery benefit, while link adaptation can affect IMPORTANT and overall delivery under independent copy loss at additional communication cost.
4. A reproducible simulation and 96-run real-device execution study that retains negative results, an unresolved later anomaly, raw records, and provenance.

Our principal finding is limited: event-aware allocation targets more communication toward critical samples and sometimes modestly improves their delivery over blind equal-budget allocation under independent copy loss. For correctly classified CRITICAL samples, the policy cap leaves link adaptation no additional copy-allocation action. Under RANDOM_COPY, the same adaptation showed higher IMPORTANT and overall delivery than IMPORTANCE-ONLY at greater communication cost.

2 Related Work

2.1 LPWAN reliability and replication

LoRaWAN specifies confirmed uplink traffic and associated downlink behavior [1], but ACKs and retransmissions incur capacity and duty-cycle costs. Capuzzo *et al.* analyze pitfalls of confirmed traffic [2], while Adelantado *et al.* describe broader LoRaWAN operating limits [3] and Magrin *et al.* examine how parameter choices, including confirmed-traffic settings, affect system performance [4]. These studies motivate cost-aware reliability evaluation; they do not provide a direct protocol baseline because our E220 devices use a custom application protocol rather than LoRaWAN. Hybrid coded replication in LoRa networks explicitly trades redundancy against reliability [5]. Our proactive copies are uncoded and deliberately simple, making the allocation rule and its cost visible. The constructed BURST_SAMPLE model further tests the limit of repeating copies within a single logical sample; it is not a measurement of LoRaWAN or RF burst errors.

2.2 Link-quality-aware adaptation

Link estimation in sensor networks has long combined observations across protocol layers, including packet and ACK outcomes [6]. Srinivasan *et al.* show that wireless losses can be bursty, so a single mean delivery rate need not characterize temporal behavior [7]. LoRaWAN adaptive-data-rate work uses link observations to adjust radio parameters and studies convergence under changing conditions [8]. Our estimator is narrower: it uses accepted first-copy ACK history to select a bounded number of application DATA copies, without changing physical-layer settings. These references support examining when history predicts future outcomes; they do not establish that this estimator captures time-varying E220 channel quality under the software-injected losses studied here.

2.3 Importance-aware and event-aware communication

Data-importance-aware retransmission for edge learning allocates radio resources according to training-sample utility as well as channel conditions [9]. Importance-aware user scheduling likewise incorporates learning-task value into wireless acquisition [10]. Broader semantic and task-oriented communication research asks how transmission should reflect the receiver’s intended task rather than treating every bit as equally valuable [11]. Event-triggered sensor-network transmission has also been studied to reduce communication devoted to measurements that do not warrant an update [12]. These works motivate application-aware prioritization, but their objectives, importance measures, and actions differ from ours: EVENTGUARD uses a fixed synthetic sensor-event classifier and proactive whole-frame replication, not model-uncertainty ARQ, semantic encoding, or event-triggered omission of routine samples. We claim neither field validation for the four sensor modalities nor priority over those methods.

2.4 Position of this study

The methodological question here is narrower than a general claim for adaptive LPWAN reliability. We separate predicted event importance from ACK-history link response, compare allocation at an exact DATA-copy budget, and test both independent copy erasures and common-mode sample erasures. The real-device study checks execution of this frozen logic and its post-reception injection path on one E220 pair. It cannot adjudicate the relative performance of the cited methods on their own protocols, workloads, or measured radio channels.

3 System Model and Problem Formulation

A **logical sample** i is one timestamped vector x_i of temperature, humidity, light, and soil-moisture values. The synthetic trace generator also assigns a ground-truth event category $g_i \in \{N, I, C\}$ from phase semantics and hazard criteria; the on-node classifier does not read g_i . It predicts $\hat{g}_i$ from sensor values and its prior state. A **physical DATA copy** is one transmitted frame for sample i , identified by the same logical sequence number and a copy index. The gateway validates its CRC, applies any planned receive-side DATA erasure, and deduplicates copies before counting a logical delivery. Accepted DATA elicits an ACK frame. An ACK may be physically received yet rejected by a planned application-layer ACK erasure; only an accepted first-copy ACK enters the link estimator. A scheduled set of copies is proactive redundancy, not an ACK-timeout retransmission.

Let $r_i \in \{1, 2, 3\}$ be the selected number of DATA copies and $Y_i = 1$ when at least one copy of sample i is accepted after the specified injected-loss path; otherwise $Y_i = 0$. Let $\mathcal{C} = \{i : g_i = C\}$. The primary reliability metric is

$$R_C = \frac{\sum_{i \in \mathcal{C}} Y_i}{|\mathcal{C}|}, \quad B = \sum_i r_i. \quad (1)$$

At a fixed DATA-copy budget B , the motivating allocation problem is to improve R_C , while tracking physical DATA copies, DATA/ACK bytes, and other communication costs. EVENTGUARD is a deterministic heuristic for this problem, not a claimed optimizer. Equal B does **not** imply equal total bytes because delivered copies generate ACK activity. Matching additionally holds the seed-specific trace and canonical DATA/ACK loss calendar fixed. The first-copy calendar is shared across strategies; effective loss over attempted copies can differ when strategies select different copy opportunities.

4 EventGuard Design

4.1 Importance estimation

The frozen classifier uses normalized inter-sample change, deviation from a normal-state baseline, rate of change, simultaneous change across channels, and event persistence. The four normalization scales are 0.5 °C, 5 humidity points, 100 lux, and 8 soil-moisture points. A directional risk rule identifies severe soil-moisture decline or combined elevated temperature/humidity.

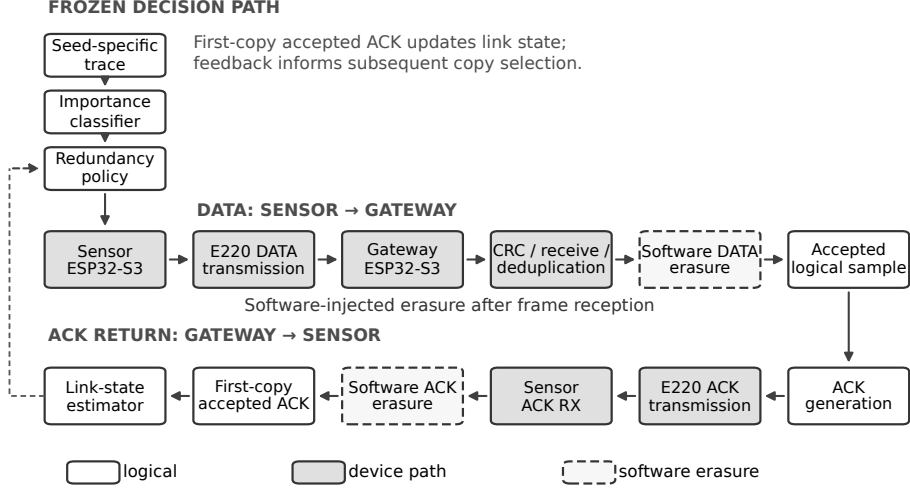


Figure 1: Frozen v1 device and decision paths. The seed-specific trace drives classification and copy selection on the Sensor ESP32-S3. The main DATA path runs left to right from Sensor through E220 and Gateway reception to the accepted logical sample; ACK follows the return path. Dashed boxes mark software erasures applied after frame reception. Only accepted first-copy ACK outcomes update the link-state estimator and inform subsequent copy selection. Configured loss is not a measured RF packet-error rate.

The IMPORTANT score threshold is 0.85 and the CRITICAL directional-risk threshold is 3.0. Normal observations update the baseline with coefficient 0.06. Exact equations and evaluation hashes are preserved in the frozen algorithm specification. The synthetic-trace classifier’s CRITICAL precision (0.8125) and recall (1.0000) describe this constructed benchmark; they are not field accuracy estimates.

4.2 First-copy link-state estimation

The estimator records accepted-ACK outcomes for the **first** DATA copy of each logical sample, avoiding the strategy-dependent observation count that would result from including every extra copy. A 12-observation sliding window gives a first-copy success ratio. State is BAD for three consecutive first-copy failures or a ratio below 0.50; otherwise DEGRADED below 0.80; otherwise GOOD. The initial no-observation ratio is one. Physical ACK reception, injected ACK drop, and accepted ACK remain separate records.

Table 1: Frozen v1 parameters relevant to the primary evaluation. Study matrix fields are taken from the frozen evaluation manifest; the root default configuration also contains older pilot-matrix defaults, discussed in [Section 6.9](#).

Parameter	Value
Trace	9 phases $\times$ 6 samples = 54 samples per seed
Inter-sample interval	10 s
IMPORTANT / CRITICAL threshold	0.85 / 3.0
Importance baseline coefficient	0.06
Link window	12 first-copy outcomes
DEGRADED / BAD ratio thresholds	0.80 / 0.50
BAD consecutive first-copy failures	3
Maximum copies per sample	3
ACK wait timeout	1,000 ms
DATA / ACK frame length	26 / 13 bytes
E220 UART rate	9,600 bit/s

4.3 Redundancy allocation

Table 1 records the frozen parameters. The base mapping assigns NORMAL/IMPORTANT/CRITICAL to one/two/three copies on a GOOD link. DEGRADED adds one base copy, clipped to the three-copy cap. BAD adds two only to IMPORTANT or CRITICAL, likewise clipped; NORMAL remains at one. The complete decision map is shown in Fig. 2. All selected copies are transmitted irrespective of an earlier ACK.

4.4 Structural implication

For a **correctly classified** CRITICAL sample, $r_i = 3$ even when the link is GOOD. Because $r_i \leq 3$, no DEGRADED or BAD transition can raise its redundancy. This is a constraint of frozen v1, not an empirical estimate. Link adaptation can alter NORMAL and IMPORTANT copy counts, and might matter for a misclassified critical event; on the evaluated synthetic trace, however, CRITICAL recall was 1.0000. A critical-delivery advantage over IMPORTANCE-ONLY therefore had little policy action space in this experiment.

	GOOD	DEGRADED	BAD
NORMAL	1 DATA copies	2 DATA copies	1 DATA copies
IMPORTANT	2 DATA copies	3 DATA copies	3 DATA copies
CRITICAL	3 DATA copies	3 DATA copies	3 DATA copies

CRITICAL has no remaining link-adaptive action at the three-copy cap.

Figure 2: Frozen v1 redundancy decision matrix. Cell values are proactive DATA copies per logical sample, indexed by predicted importance and pre-sample link state. The CRITICAL row remains 3/3/3 at the frozen three-copy cap; this is policy structure, not an empirical estimate.

5 Baselines and Budget Fairness

Table 2 separates diagnostic ablations from exact-budget comparisons. The fixed rules, FIXED-1, FIXED-2, and FIXED-3, send a constant one, two, or three copies. IMPORTANCE-ONLY applies the frozen 1/2/3 importance rule without link adaptation. LINK-ONLY applies 1/2/3 copies for GOOD, DEGRADED, and BAD, respectively, without importance. Neither fixed baselines nor component ablations are equal-cost by construction, so their delivery contrasts must be read alongside cost.

For each matched model/rate/seed/trace, the EVENTGUARD reference determines B . UNIFORM-BUDGET first assigns one copy to each sample, then allocates remaining copies by a fixed round-robin order up to the cap. RANDOM-BUDGET allocates the extras by a seeded hash ordering. Their decisions may use the sample index, fixed budget, and allocation seed, but not ground truth, predicted importance, classifier score, or sensor values. The hardware budget is computed from the frozen host reference before strategy execution, allowing a seeded, interleaved execution order rather than always running EVENTGUARD first. Actual DATA-copy totals are checked against the reference budget.

Table 2: Evaluated strategies and comparison roles. All eight appear in the host matrix; the four marked hardware appear in the final 96-run set.

Strategy	Copy rule	Role	Hardware set
FIXED-1	One per sample	Minimal fixed-copy comparison	No
FIXED-2	Two per sample	Fixed-copy comparison	No
FIXED-3	Three per sample	Maximum fixed-copy comparison	No
IMPORTANCE-ONLY	Predicted importance, 1/2/3	Importance and link ablation	Yes
LINK-ONLY	GOOD/DEGRADED/ BAD, 1/2/3	Link-only ablation	No
EVENTGUARD	Importance plus first-copy link state, cap 3	Combined treatment	Yes
UNIFORM-BUDGET	Event-blind round-robin, exact B	Equal-budget control	Yes
RANDOM-BUDGET	Event-blind seeded order, exact B	Equal-budget control	Yes

6 Experimental Methodology

6.1 Synthetic trace and labels

Each evaluation seed generates one deterministic 54-sample realization with nine phases and a 10-second sampling interval. Small seed-dependent sensor noise makes trace hashes differ across seeds. All strategies within a paired seed receive the same realization. Values cover temperature, humidity, light, and soil moisture; benchmark runs replay these values rather than sampling the attached physical sensors. Ground-truth categories derive from the generator’s phase semantics and hazard conditions. Predicted importance derives only from observed values and classifier history. Calibration seeds 1–30 were kept separate from evaluation seeds 31–130.

6.2 Injected-loss models

RANDOM_COPY makes deterministic, independently keyed DATA and ACK erasure decisions for canonical copy opportunities. BURST_SAMPLE marks contiguous logical samples and erases all three canonical copy opportunities of an affected sample. DATA and ACK calendars are separate and

shared across strategies at a fixed model, rate, and seed. By construction, extra copies within a `BURST_SAMPLE`-erased sample cannot recover that sample. The model isolates common-mode failure; it does not represent every physical burst-fading process. Configured percentages are application-layer injection settings, not measured E220 packet-error rates. The host audit retains realized fairness alarms rather than discarding unusual calendars.

6.3 Host simulation

The frozen pre-hardware matrix contains eight strategies, five configured rates (0%, 5%, 10%, 20%, 30%), two principal loss models, and 100 evaluation seeds (31–130): **8,000 runs**. Each uses its seed-specific trace and matched canonical calendars. Python/C parity checks cover importance decisions, copy selection, link transitions, fault decisions, and budget allocation. An earlier 300-run E220 pilot used a different 18-sample trace and is not pooled with frozen v1.

6.4 Sensitivity and ablation

An additional **11,000 host-only** runs vary one axis at a time: CRITICAL threshold {2.0, 2.5, 3.0, 3.5, 4.0}, link-window length {8, 12, 16, 24}, or maximum copies {2, 3}. These diagnostic perturbations do not retune the default experiment. Four copies are unsupported because v1 defines only three canonical copy opportunities; a fourth against the same calendar would receive an invalid, unerased slot. Component ablations compare FIXED-2, IMPORTANCE-ONLY, LINK-ONLY, and EVENTGUARD; exact-budget controls address allocation fairness.

6.5 Hardware execution

Two ESP32-S3 boards carried Sensor and Gateway firmware through E220-400T22D modules under ESP-IDF v5.4.4. DATA and ACK were real transmitted frames on this single device pair, using 9,600-bit/s UART. The Sensor replayed the deterministic trace; attached environmental sensors did not generate benchmark data. The Gateway received, CRC-checked, and deduplicated DATA. Planned software erasures acted after receiver-side frame arrival. Raw Sensor/Gateway serial logs, END counters, UART diagnostics, and per-run manifests were retained. A v2 12-run smoke validation preceded Stage 1. Receive-parser and host-validation repairs were engineering changes; the frozen classifier, link estimator, redundancy policy, trace, and loss calendar were not changed for these results.

Table 3: Hardware execution scope. This is a single validated path, not a general deployment.

Item	Observed setup
Nodes and topology	One Sensor ESP32-S3 and one Gateway ESP32-S3; one link
Radio modules	One E220-400T22D per node
Firmware environment	ESP-IDF v5.4.4; frozen EVENTGUARD-v1 research logic
Serial/radio path	E220 UART at 9,600 bit/s; real DATA and ACK frames
Benchmark input	Seed-specific deterministic trace replay, 54 samples/run
Programmed impairments	DATA/ACK software erasures after reception
Preserved evidence	Raw logs, run manifests, firmware hashes, counters, diagnostics

6.6 Construction of the balanced hardware analysis

The original Stage 1 plan contained four strategies $\times$ two models $\times$ two rates (20%, 30%) $\times$ ten seeds (31–40), or **160 runs**, with interleaved seeded strategy order.

During later expansion, run103 (UNIFORM-BUDGET, RANDOM_COPY, 30%, seed37) stalled while waiting for an ACK; one DATA frame was absent from Gateway pre-injection logs. Its low-level cause remains unresolved. The final set is a **post-hoc balanced analysis of the interrupted Stage 1 experiment**: the earliest contiguous seed prefix, 31–36, for which all four strategies and all four conditions were complete and auditable. This yields 96 completed runs and six paired seed units per condition. Selection used execution order, completeness, and audibility, not delivery outcomes. The sample size $n = 6$ was **not preregistered**. Partial seed-37 records and run103 remain preserved outside the paired analysis.

6.7 Integrity and reference checks

The offline finalizer re-parses preserved raw logs and checks raw/manifest hashes, firmware identities, run configuration, trace/calendar hashes, all 54 event/sample records, per-copy DATA/ACK decisions, importance and selected-copy sequences, link transitions, Sensor/Gateway END counters, UART diagnostics, and exact budget totals. All **96/96** selected runs passed. There were zero uncontrolled physical DATA misses, zero uncontrolled phys-

ical ACK misses, zero policy divergences, and zero sample-level delivery differences from matched host references. This establishes agreement for deterministic replay and software injection, not a measured RF error model. Independent USB reader byte-count telemetry absent from the historical raw schema cannot be reconstructed; firmware UART diagnostics and event lines were cross-checked instead.

6.8 Metrics and statistics

Critical-event and important-event delivery are the delivered fractions of ground-truth CRITICAL and IMPORTANT samples, respectively; overall delivery is the delivered fraction of all logical samples. Physical DATA copies count every proactive frame transmission; bytes include DATA and ACK transmissions. Critical traffic share is the fraction of DATA copies assigned to ground-truth CRITICAL samples, computed only after the run. Truth is never an allocator input. The communication-time field is the UART serialization proxy $(\text{DATA bytes} + \text{ACK bytes}) \times 10 / 9600$ seconds, not measured RF airtime. There is no measured energy or Joule estimate.

The statistical unit is one paired seed/run under a fixed condition, not a packet, copy, or logical sample. Hardware summaries report mean, median, sample standard deviation, and a two-sided 95% Student-t interval across six seed units. Paired tables also report wins/ties/losses, mean and median paired differences, and a two-sided exact conditional sign-randomization p-value for Wilcoxon signed ranks. Differences of magnitude at most $1\text{e-}12$ are treated as zero and omitted from ranking; equal absolute nonzero differences receive midranks. The exact null distribution enumerates sign assignments conditional on these midranks. When every pair ties, the implementation reports $p=1$ and rank-biserial effect size zero; otherwise rank-biserial is the positive-minus-negative signed-rank sum divided by the total rank sum. Holm-adjusted values cover the family of 12 critical-delivery contrasts (three baselines across four conditions); other metric p-values remain unadjusted. The unbounded t intervals may extend beyond $[0,1]$ for a delivery ratio and are not clipped. Given post-hoc subset construction and only six paired seeds, **all p-values and intervals are descriptive/exploratory; no confirmatory significance claim is made.**

6.9 Versioned configuration and reproducibility

The frozen specification, study manifests, source hashes, archived firmware images, raw logs, and finalizer provenance identify the evaluated version.

The root default configuration retains older pilot-oriented matrix defaults; the frozen-v1 study manifests and recorded calls define the 54-sample trace, evaluated strategies, and loss models used here. Reproduction should follow those versioned records rather than infer the completed experiment from the root defaults.

7 Results

7.1 Host-simulation overview

In the 100-seed host matrix, EVENTGUARD and IMPORTANCE-ONLY had identical CRITICAL delivery in every matched run. Under RANDOM_COPY at 20% configured loss, their mean CRITICAL delivery was 0.9892, versus 0.9746 for equal-budget UNIFORM-BUDGET and 0.9769 for equal-budget RANDOM-BUDGET. At 30%, the corresponding means were 0.9723, 0.9438, and 0.9492. Directing the same total copy budget by predicted importance can help under independent copy erasures; this does not establish a gain from the link estimator. At RANDOM_COPY 20%, EVENTGUARD versus IMPORTANCE-ONLY mean IMPORTANT delivery was 0.9841 versus 0.9572 and mean overall delivery was 0.9596 versus 0.9298; their mean DATA copies were 140.68 versus 112.00. At 30%, IMPORTANT delivery was 0.9748 versus 0.9124, overall delivery 0.9333 versus 0.8822, and copies 142.65 versus 112.00. These 100-seed host means show a secondary-outcome gain at increased cost while CRITICAL delivery remains equal. At 0% loss, delivery saturated at 1.0000 for all strategies. Under BURST_SAMPLE, all strategies tied in CRITICAL delivery within each rate.

7.2 Ablation

The host ablation separates importance from link response. At RANDOM_COPY 20%, FIXED-2 delivered 0.9600 of CRITICAL events, versus 0.9892 for both IMPORTANCE-ONLY and EVENTGUARD; their mean DATA-copy counts were 108.00, 112.00, and 140.68. At 30%, the three CRITICAL delivery means were 0.9185, 0.9723, and 0.9723; EVENTGUARD used 142.65 copies versus 112.00 for IMPORTANCE-ONLY. LINK-ONLY delivered 0.942 at 20% and 0.938 at 30% in the same host model. These component comparisons are **not equal-budget tests**. Importance, rather than added link response, accounts for the combined policy’s observed CRITICAL-delivery level. The IMPORTANT and overall curves in Fig. 4 reveal the link response’s remaining action on other classes; these gains require additional copies.

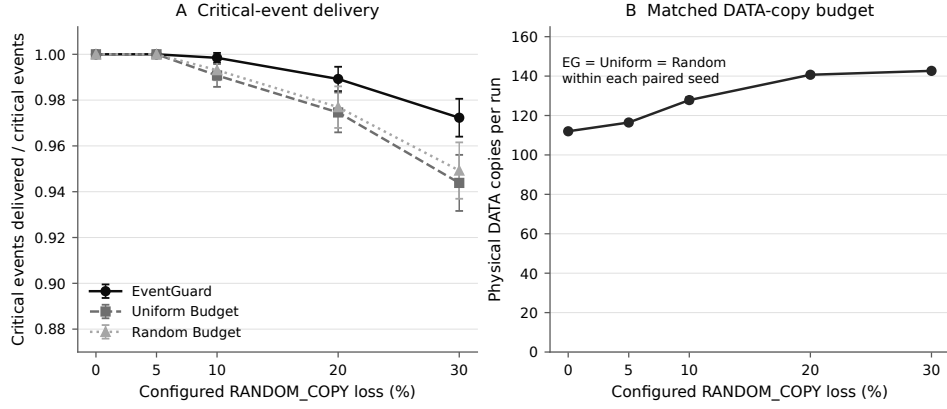


Figure 3: Exact-budget host comparison under software-injected `RANDOM_COPY` loss. Panel A shows mean critical-event delivery ratios (vertical axis displayed from 0.87 to 1.01); Panel B shows mean physical DATA copies per 54-sample run. Error bars are two-sided 95% Student-t intervals across 100 paired evaluation seeds (31–130). `EVENTGUARD`, `UNIFORM-BUDGET`, and `RANDOM-BUDGET` have exactly matched DATA-copy counts within each seed and configured loss rate (0%, 5%, 10%, 20%, 30%); the single Panel B series represents all three.

7.3 Sensitivity

Across link-window lengths 8/12/16/24, mean `CRITICAL` delivery did not change within the reported host conditions, although mean copies and state transitions varied. The `CRITICAL`-threshold sweep from 2.0 to 4.0 likewise changed copy allocation slightly without changing reported condition-level `CRITICAL` delivery. Increasing the copy cap from two to three improved host `CRITICAL` delivery under `RANDOM_COPY` (0.9600 to 0.9892 at 20%; 0.9185 to 0.9723 at 30%) at greater cost. Under `BURST_SAMPLE` at 20% or 30%, the cap change left `CRITICAL` delivery unchanged. No cap-four result exists in frozen v1.

The associated sensitivity visualization is presented as Fig. 6 after the hardware cost figure.

7.4 Real-device execution results

Table 4 reports means over the six paired hardware seeds. Its displayed ratio is the fraction of ground-truth `CRITICAL` samples delivered; `IMPORTANT`

Table 4: Final 96-run balanced hardware summary (six paired seeds per condition). RC denotes `RANDOM_COPY`; BS denotes `BURST_SAMPLE`. Critical traffic share uses truth only in retrospective analysis.

Condition	Strategy	Critical delivery	DATA copies	Total bytes	Critical traffic (%)
RC 20%	EVENTGUARD	0.9744	142.50	5,154.5	27.4
RC 20%	IMPORTANCE-ONLY	0.9744	112.00	4,056.0	34.8
RC 20%	UNIFORM-BUDGET	0.9744	142.50	5,165.3	23.0
RC 20%	RANDOM-BUDGET	0.9615	142.50	5,167.5	24.1
RC 30%	EVENTGUARD	0.9487	145.00	5,033.2	26.9
RC 30%	IMPORTANCE-ONLY	0.9487	112.00	3,913.0	34.8
RC 30%	UNIFORM-BUDGET	0.9231	145.00	5,057.0	24.0
RC 30%	RANDOM-BUDGET	0.9231	145.00	5,063.5	24.0
BS 20%	EVENTGUARD	0.7821	133.17	4,857.7	29.3
BS 20%	IMPORTANCE-ONLY	0.7821	112.00	4,084.2	34.8
BS 20%	UNIFORM-BUDGET	0.7821	133.17	4,862.0	21.1
BS 20%	RANDOM-BUDGET	0.7821	133.17	4,851.2	23.9
BS 30%	EVENTGUARD	0.7179	139.17	4,894.5	28.1
BS 30%	IMPORTANCE-ONLY	0.7179	112.00	3,943.3	34.8
BS 30%	UNIFORM-BUDGET	0.7179	139.17	4,920.5	22.7
BS 30%	RANDOM-BUDGET	0.7179	139.17	4,894.5	23.9

and overall ratios are reported immediately below, while copies and bytes are per-run means. Median, standard deviation, confidence interval, and paired statistics remain in the preserved final analysis files. All configured losses were injected in software after reception.

The selected hardware `RANDOM_COPY` runs reveal the secondary delivery objectives. At 20%, `EVENTGUARD` and `IMPORTANCE-ONLY` both delivered 0.9744 of `CRITICAL` events, while `IMPORTANT` delivery was 0.9885 versus 0.9540 and overall delivery 0.9599 versus 0.9259. `EVENTGUARD` used 142.50 versus 112.00 `DATA` copies and 5,154.5 versus 4,056.0 total bytes per run. At 30%, both strategies delivered 0.9487 of `CRITICAL` events, while `IMPORTANT` delivery was 0.9713 versus 0.9253 and overall delivery 0.9259 versus 0.8920; `EVENTGUARD` used 145.00 versus 112.00 copies and 5,033.2 versus 3,913.0 bytes. Thus the observed `EVENTGUARD`-minus-`IMPORTANCE-ONLY` `IMPORTANT` gains were 0.0345 and 0.0460, and overall gains were 0.0340 at both rates, with 30.5 and 33.0 extra copies. Under `BURST_SAMPLE` at both rates, the two strategies had identical `CRITICAL`, `IMPORTANT`, and overall delivery while `EVENTGUARD` used more copies. These are descriptive means across six paired seeds, not confirmatory effect estimates.

All 96 selected runs passed offline checks and showed zero sample-level delivery differences from deterministic host references. This supports faithful execution of the frozen trace, policy, and injected calendar on the device path, not prediction of an uncontrolled radio channel.

7.5 Exact-budget paired comparisons

UNIFORM-BUDGET and RANDOM-BUDGET used exactly EVENTGUARD’s DATA-copy count in every selected condition and seed. EVENTGUARD allocated a larger share of copies to ground-truth CRITICAL samples than either blind allocator in all four conditions (Table 4). Under RANDOM_COPY 30%, EVENTGUARD exceeded each blind baseline in **two of six** paired seeds and tied four; its mean paired critical-delivery difference was +0.0256 against each. The unadjusted exact Wilcoxon p-value was 0.50 for each, and the exploratory Holm-adjusted value across 12 critical-delivery comparisons was 1.00. Under RANDOM_COPY 20%, it tied UNIFORM-BUDGET in all six pairs and exceeded RANDOM-BUDGET in one, with a +0.0128 mean difference. Both BURST_SAMPLE rates yielded six ties against each blind baseline. These sparse differences support targeted allocation and limited condition-specific delivery gains, not confirmatory general superiority. Equal DATA-copy budgets need not produce identical total bytes because ACK activity varies.

7.6 Pareto tradeoff

IMPORTANCE-ONLY and EVENTGUARD had **identical CRITICAL delivery in all 24 matched hardware seed/condition pairs**. EVENTGUARD nevertheless used 30.50, 33.00, 21.17, and 27.17 additional DATA copies per run on average in RC 20%, RC 30%, BS 20%, and BS 30%. Extra total bytes were 1,098.5, 1,120.2, 773.5, and 951.2. For the evaluated CRITICAL-delivery/cost objective, on the empirical within-condition frontier of mean CRITICAL delivery versus DATA copies or bytes, IMPORTANCE-ONLY appeared in **4/4** conditions and EVENTGUARD in **0/4**. These are descriptive frontiers over observed means, not uncertainty regions.

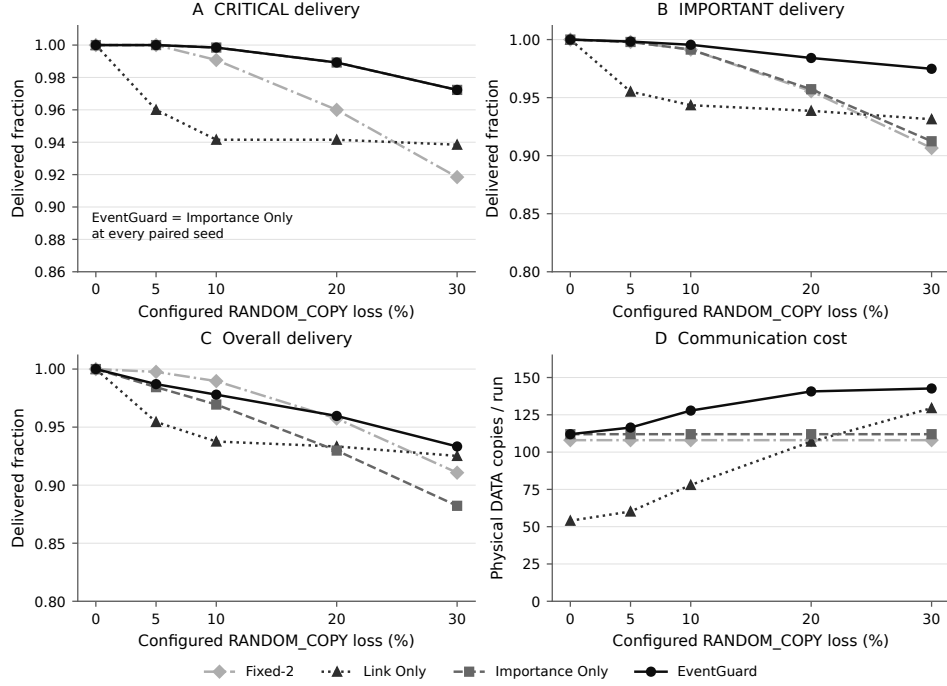


Figure 4: Host ablation under software-injected `RANDOM_COPY` loss. Panels A, B, and C plot mean CRITICAL, IMPORTANT, and overall delivery; Panel D plots mean physical DATA copies per 54-sample run. Each point averages 100 paired evaluation seeds (31–130) for FIXED-2, LINK-ONLY, IMPORTANCE-ONLY, and EVENTGUARD. EVENTGUARD and IMPORTANCE-ONLY coincide on CRITICAL delivery at every matched seed; their IMPORTANT and overall outcomes and DATA-copy costs can differ. These component ablations are not equal-budget comparisons.

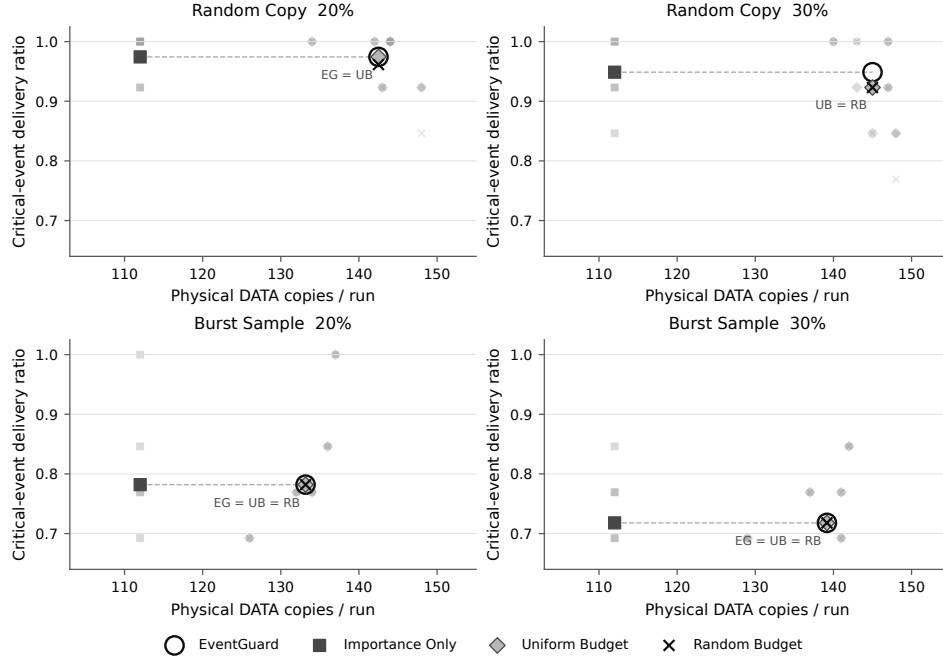


Figure 5: Real-device critical-event delivery versus DATA-copy cost. Each panel is one configured software-injected loss model/rate; small faint markers show the six preserved seed-level runs (31–36) per strategy, while larger markers show condition means of critical-event delivery and physical DATA copies per 54-sample run. Seed-level points are plotted without jitter. The delivery axis is displayed from 0.64 to 1.025. The 96-run set is a post-hoc balanced prefix from one ESP32-S3/E220 device pair. The dashed horizontal segment connects IMPORTANCE-ONLY and EVENTGUARD at identical observed mean delivery but different mean copy cost. Coincident event-blind mean points share their exact plotted coordinates. This empirical Pareto display is descriptive, not a significance test or measured RF-loss result.

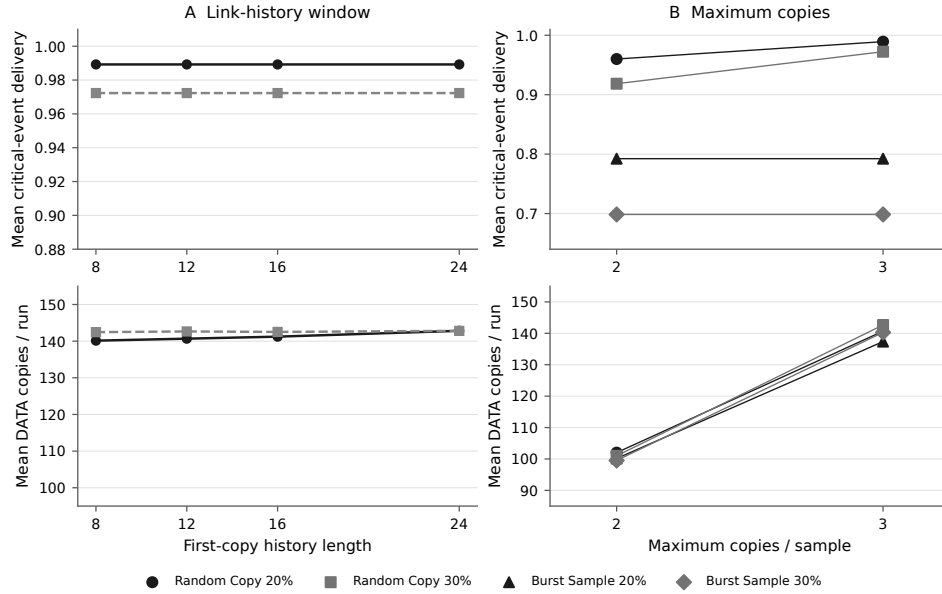


Figure 6: Frozen host-only sensitivity under software-injected loss. Left: mean critical-event delivery and physical DATA copies per 54-sample run for first-copy link-history windows 8, 12, 16, and 24 under `RANDOM_COPY` at 20% and 30%. Right: the same metrics for maximum redundancy 2 versus 3 under `RANDOM_COPY` and `BURST_SAMPLE` at 20% and 30%; thin straight segments connect only the two evaluated cap settings within each condition. Each plotted mean uses 100 evaluation seeds; the segments do not imply intermediate evaluations, and no cap-four result is shown.

8 KPI-Dependent Effects of Link Adaptation

8.1 Structural saturation for CRITICAL

The frozen CRITICAL row is 3/3/3 across GOOD, DEGRADED, and BAD, equal to the three-copy cap. Correctly classified CRITICAL samples therefore have no remaining link-adaptive copy-allocation action. Synthetic-benchmark CRITICAL recall was 1.0000, limiting the route through misclassification in these traces without implying field accuracy. EVENTGUARD and IMPORTANCE-ONLY had identical CRITICAL delivery in every matched host run and all selected hardware pairs. This equality is largely expected from the frozen policy structure, not evidence that link adaptation is universally ineffective or that a statistical test established equivalence.

8.2 Remaining action space for IMPORTANT and NORMAL

Link state can still change selected copies for IMPORTANT and NORMAL samples. Under independently keyed RANDOM_COPY erasures, the additional copies coincide with higher IMPORTANT and overall delivery in host and selected device data (Sections 7.1 and 7.4). A CRITICAL-only headline metric would miss these observed secondary outcomes. This is an action-space and KPI distinction, not a claim that accepted first-copy ACK history predicts measured E220 fading.

8.3 Communication-cost tradeoff

The secondary gains are not free. In selected hardware RANDOM_COPY runs, EVENTGUARD used 30.5 extra DATA copies and 1,098.5 extra total bytes per run at 20%, and 33.0 extra copies and 1,120.2 extra bytes at 30%, relative to IMPORTANCE-ONLY. For the evaluated CRITICAL-delivery objective, IMPORTANCE-ONLY achieved the same outcome at lower cost and occupied the corresponding empirical frontier; that ranking does not extend automatically to IMPORTANT or overall delivery. The useful design question is which KPI gains justify added communication expenditure.

8.4 Dependence on programmed loss structure

RANDOM_COPY erasures are keyed separately by copy opportunity, so extra copies can protect non-saturated classes. BURST_SAMPLE erases every canonical copy opportunity for an affected logical sample; repeating that sample

cannot reverse its programmed erasure. This construction explains the selected hardware ties across all three delivery KPIs under `BURST_SAMPLE`. It does not establish how a real burst-fading, interference, or range process would respond to redundancy.

9 Discussion

The evaluation separates allocation from adaptation and exposes a KPI-dependent result. Event importance determines where an exact DATA-copy budget is spent, and `EVENTGUARD` directed more of that budget to `CRITICAL` samples than event-blind controls. Its `CRITICAL`-delivery advantage over those controls was small and condition-dependent under `RANDOM_COPY` and absent under constructed sample-wide erasure. The importance-only ablation shows a different question: correctly classified `CRITICAL` samples already received three copies in every state, so `EVENTGUARD` could not raise their count. Its observed `CRITICAL` delivery equaled `IMPORTANCE-ONLY` while its cost rose. For that `CRITICAL`-delivery/cost objective, `IMPORTANCE-ONLY` occupied the empirical frontier in all four selected hardware conditions.

The same link response retained control authority over `IMPORTANT` and `NORMAL` samples. `EVENTGUARD` showed higher `IMPORTANT` and overall delivery than `IMPORTANCE-ONLY` under `RANDOM_COPY` in both host and selected hardware data, at greater DATA-copy and byte cost. A component can have no incremental effect for a saturated objective yet improve another objective on which it can act. A single headline reliability ratio obscures this tradeoff. These descriptive outcomes do not establish the optimal cost allocation across KPIs or a predictable physical RF channel.

The 96 selected device runs agree with matched host references at the sample-outcome and policy-decision levels, supporting faithful execution of the frozen logic and deterministic injection on one E220 path. They do not establish fading, interference tolerance, range, LoRaWAN behavior, or energy efficiency. The preserved incomplete run103 exposes an unresolved receive-path stall beyond the balanced prefix. These are distinct questions of simulator fidelity, device-path correctness, and channel performance; only the first two are addressed by the present study.

Table 5: Scope boundaries and consequences. These bounds apply to all three delivery KPIs and their cost tradeoffs.

Boundary	Consequence for interpretation
Synthetic, seed-specific 54-sample traces	Limited event diversity; no field-labeled classifier validation
One Sensor–Gateway device pair and one topology	No multi-node contention or multi-gateway diversity evidence
Deterministic post-reception DATA/ACK injection	Configured 20%/30% loss is not measured physical RF PER
No controlled fading, interference, or range sweep	No claim about uncontrolled time-varying RF performance
UART serialization proxy only	No measured RF PHY airtime, regulatory duty-cycle, or Joule energy result
Post-hoc complete prefix, six seeds per condition	Descriptive/exploratory inference; broad population claims unsupported
Unresolved seed-37 run103 stall	Hardware reliability outside the selected clean runs remains uncertain
Custom E220 application protocol	Results do not characterize LoRaWAN confirmed-message operation

10 Limitations

Seed-level replication cannot substitute for independent device pairs, antennas, placements, interference regimes, or measured channel histories. The classifier uses designed synthetic labels; its confusion matrix is not a field accuracy estimate. No multi-node traffic contention, multi-gateway reception, current draw, regulatory duty-cycle, or measured physical packet-error rate was evaluated. The earlier v0 pilot remains separate because its shorter trace and design cannot be pooled with frozen v1.

11 Future Work

A next experiment should vary an actual radio channel over time while recording physical reception, available RSSI/SNR or another link observable, state transitions, and unplanned errors separately from deliberate injection. A separately versioned EVENTGUARD-v2 design could provide genuine action space for CRITICAL events—for example, a changed copy cap paired with an expanded canonical calendar—before testing whether link prediction adds value. This requires renewed Python/C parity and device-path validation rather than reinterpretation of v1. Multi-node contention

and multi-gateway spatial redundancy would test mechanisms absent from the single-link study. Field-labeled events would assess classifier validity; measured current and RF airtime would allow energy and regulatory-cost analysis. These are new studies, not a proposal to fill the interrupted Stage 1 matrix for appearance’s sake.

12 Conclusion

Frozen EVENTGUARD-v1 combines predicted event importance and first-copy ACK history to allocate proactive DATA copies. At an exact budget it directed more copies to CRITICAL samples than blind controls, with limited, condition-dependent CRITICAL-delivery gains under independent programmed loss and none under constructed sample-wide erasure. For correctly classified CRITICAL samples, however, the three-copy policy cap left link adaptation no further allocation action: EVENTGUARD matched IMPORTANCE-ONLY on CRITICAL delivery in matched host and selected device runs while spending more. Under RANDOM_COPY, the same link-adaptive mechanism showed higher IMPORTANT and overall delivery in observed host and device data, at increased communication cost. Its value is therefore KPI-, action-space-, and loss-structure-dependent rather than universally absent.

Future designs should align adaptive control authority with the target delivery KPI and evaluate whether added communication cost is justified. Controlled, time-varying RF studies are needed before generalizing beyond deterministic post-reception injection and one device pair.

Data and Code Availability

The public EventGuard-LoRa repository and frozen v1.0.0 release provide the algorithm specification, source code, archived experimental manifests and raw logs, analysis scripts, and identifiers for the selected 96-run hardware dataset. The repository’s README and final hardware analysis files give offline reproduction commands; `tools/finalize_hardware_dataset.py` regenerates the final audit and descriptive statistics from preserved records without accessing devices. Public access supports inspection and citation. Reuse permissions are described in the repository’s NOTICE.md.

References

- [1] LoRa Alliance. TS001-1.0.4 LoRaWAN L2 1.0.4 Specification. Technical report, LoRa Alliance, 2023. Official source: LoRa Alliance specification page.
- [2] Martina Capuzzo, Davide Magrin, and Andrea Zanella. Confirmed traffic in LoRaWAN: Pitfalls and countermeasures. In *2018 17th Annual Mediterranean Ad Hoc Networking Workshop (Med-Hoc-Net)*, pages 1–7. IEEE, 2018. doi: <https://doi.org/10.23919/MedHocNet.2018.8407095>.
- [3] Ferran Adelantado, Xavier Vilajosana, Pere Tuset-Peiro, Borja Martinez, Joan Melia-Segui, and Thomas Watteyne. Understanding the limits of LoRaWAN. *IEEE Communications Magazine*, 55(9):34–40, 2017. doi: <https://doi.org/10.1109/MCOM.2017.1600613>.
- [4] Davide Magrin, Martina Capuzzo, and Andrea Zanella. A thorough study of LoRaWAN performance under different parameter settings. *IEEE Internet of Things Journal*, 7(1):116–127, 2020. doi: <https://doi.org/10.1109/JIOT.2019.2946487>.
- [5] Jean Michel de Souza Sant’Ana, Arliones Hoeller, Richard Demo Souza, Samuel Montejo-Sanchez, Hirley Alves, and Mario de Noronha-Neto. Hybrid coded replication in LoRa networks. *IEEE Transactions on Industrial Informatics*, 16(8):5577–5585, 2020. doi: <https://doi.org/10.1109/TII.2020.2966120>.
- [6] Rodrigo Fonseca, Omprakash Gnawali, Kyle Jamieson, and Philip Levis. Four-bit wireless link estimation. In *Proceedings of the Sixth Workshop on Hot Topics in Networks (HotNets VI)*, 2007. Official source: HotNets VI paper PDF.
- [7] Kannan Srinivasan, Maria A. Kazandjieva, Saatvik Agarwal, and Philip Levis. The β -factor: Measuring wireless link burstiness. In *Proceedings of the 6th ACM Conference on Embedded Networked Sensor Systems (SenSys)*, pages 29–42. ACM, 2008. doi: <https://doi.org/10.1145/1460412.1460416>.
- [8] Joseph Finnegan, Ronan Farrell, and Stephen Brown. Analysis and enhancement of the LoRaWAN adaptive data rate scheme. *IEEE Internet of Things Journal*, 7(8):7171–7180, 2020. doi: <https://doi.org/10.1109/JIOT.2020.2982745>.

- [9] Dongzhu Liu, Guangxu Zhu, Qunsong Zeng, Jun Zhang, and Kaibin Huang. Wireless data acquisition for edge learning: Data-importance aware retransmission. *IEEE Transactions on Wireless Communications*, 20(1):406–420, 2021. doi: <https://doi.org/10.1109/TWC.2020.3024980>.
- [10] Dongzhu Liu, Guangxu Zhu, Jun Zhang, and Kaibin Huang. Data-importance aware user scheduling for communication-efficient edge machine learning. *IEEE Transactions on Cognitive Communications and Networking*, 7(1):265–278, 2021. doi: <https://doi.org/10.1109/TCCN.2020.2999606>.
- [11] Qiao Lan, Dingzhu Wen, Zezhong Zhang, Qunsong Zeng, Xu Chen, Petar Popovski, and Kaibin Huang. What is semantic communication? a view on conveying meaning in the era of machine intelligence. *Journal of Communications and Information Networks*, 6(4):336–371, 2021. doi: <https://doi.org/10.23919/JCIN.2021.9663101>.
- [12] Xiaohua Ge, Qing-Long Han, and Zidong Wang. A dynamic event-triggered transmission scheme for distributed set-membership estimation over wireless sensor networks. *IEEE Transactions on Cybernetics*, 49(1):171–183, 2019. doi: <https://doi.org/10.1109/TCYB.2017.2769722>.